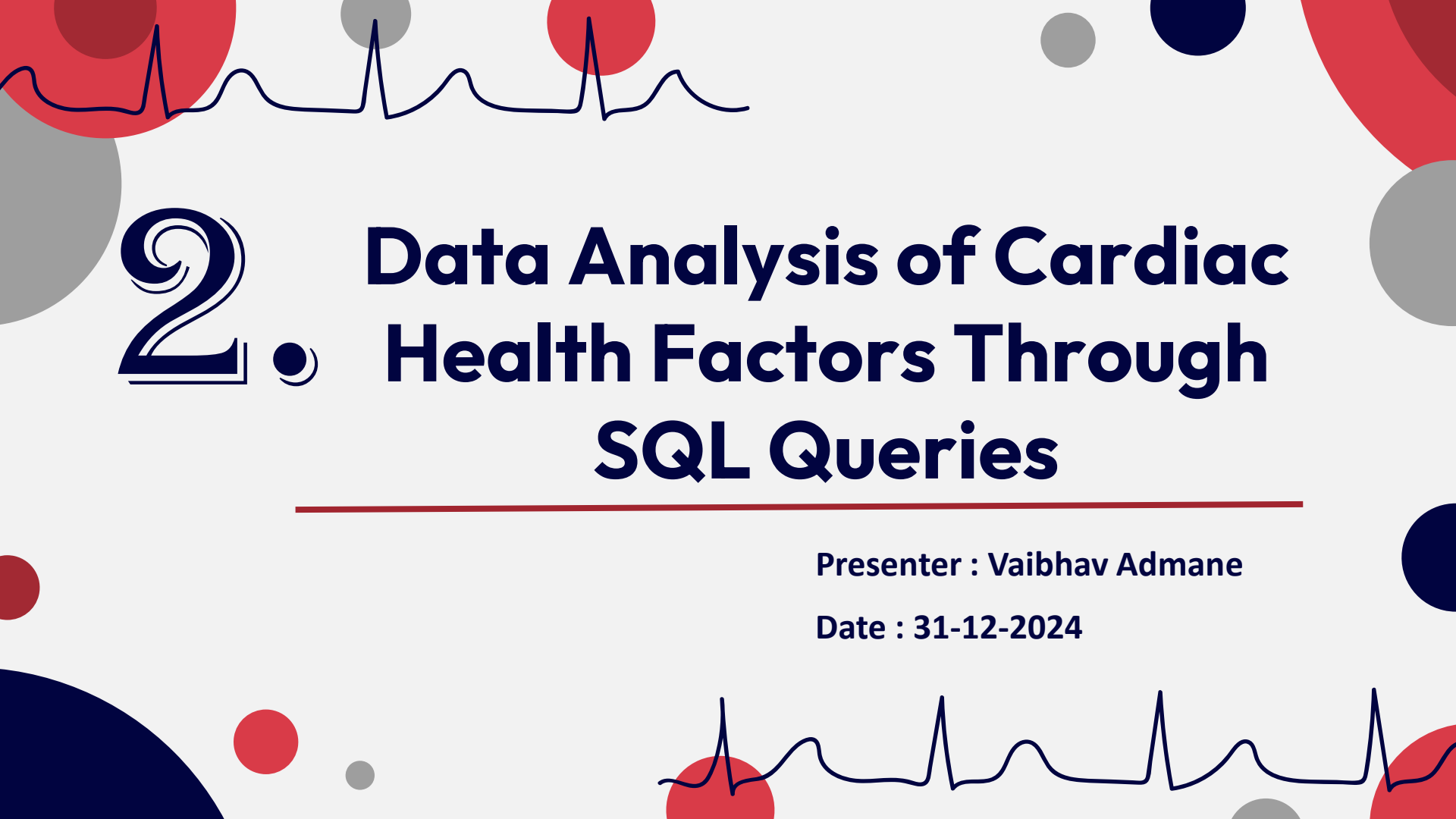




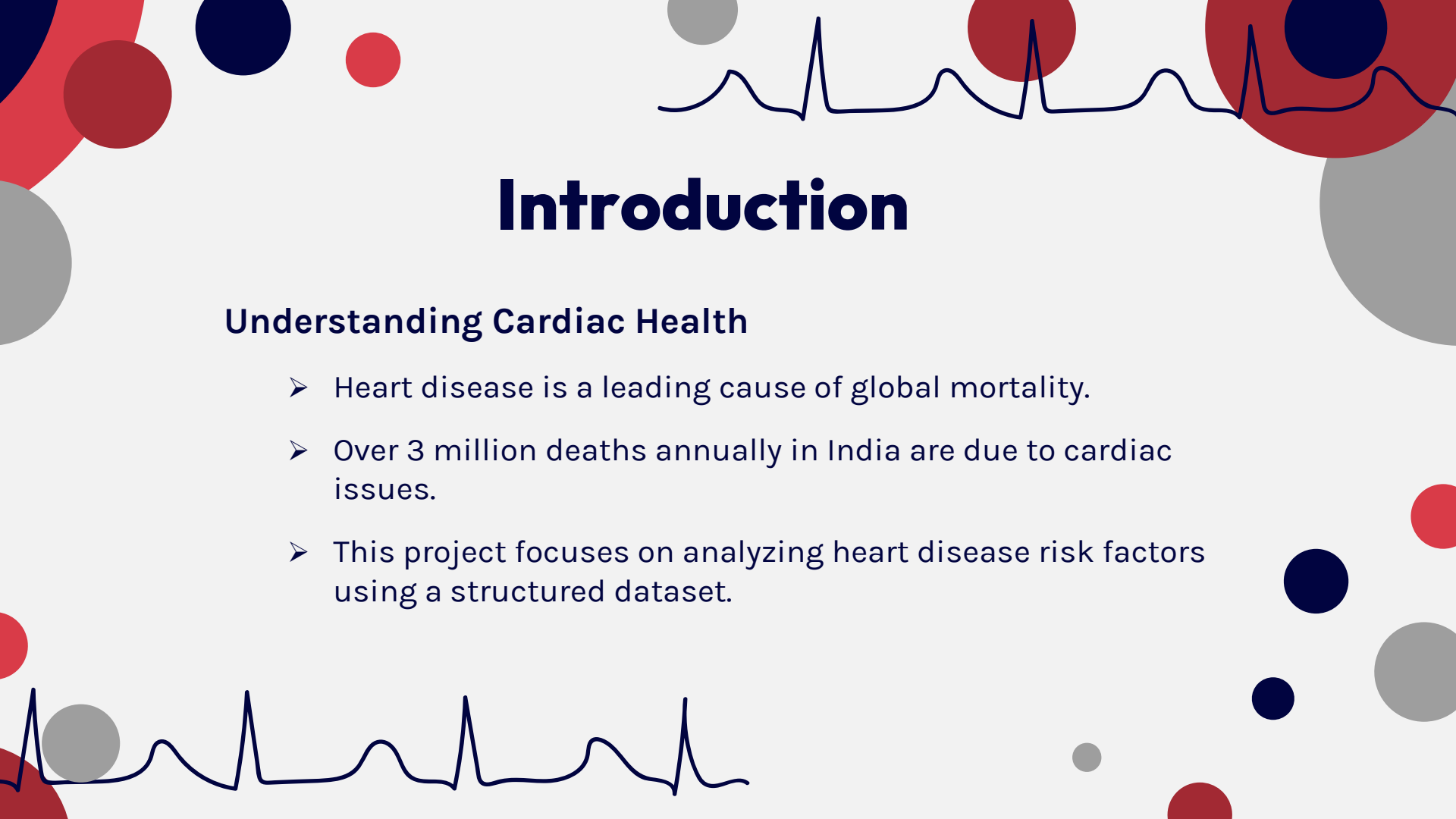
# 2

# **Data Analysis of Cardiac Health Factors Through SQL Queries**

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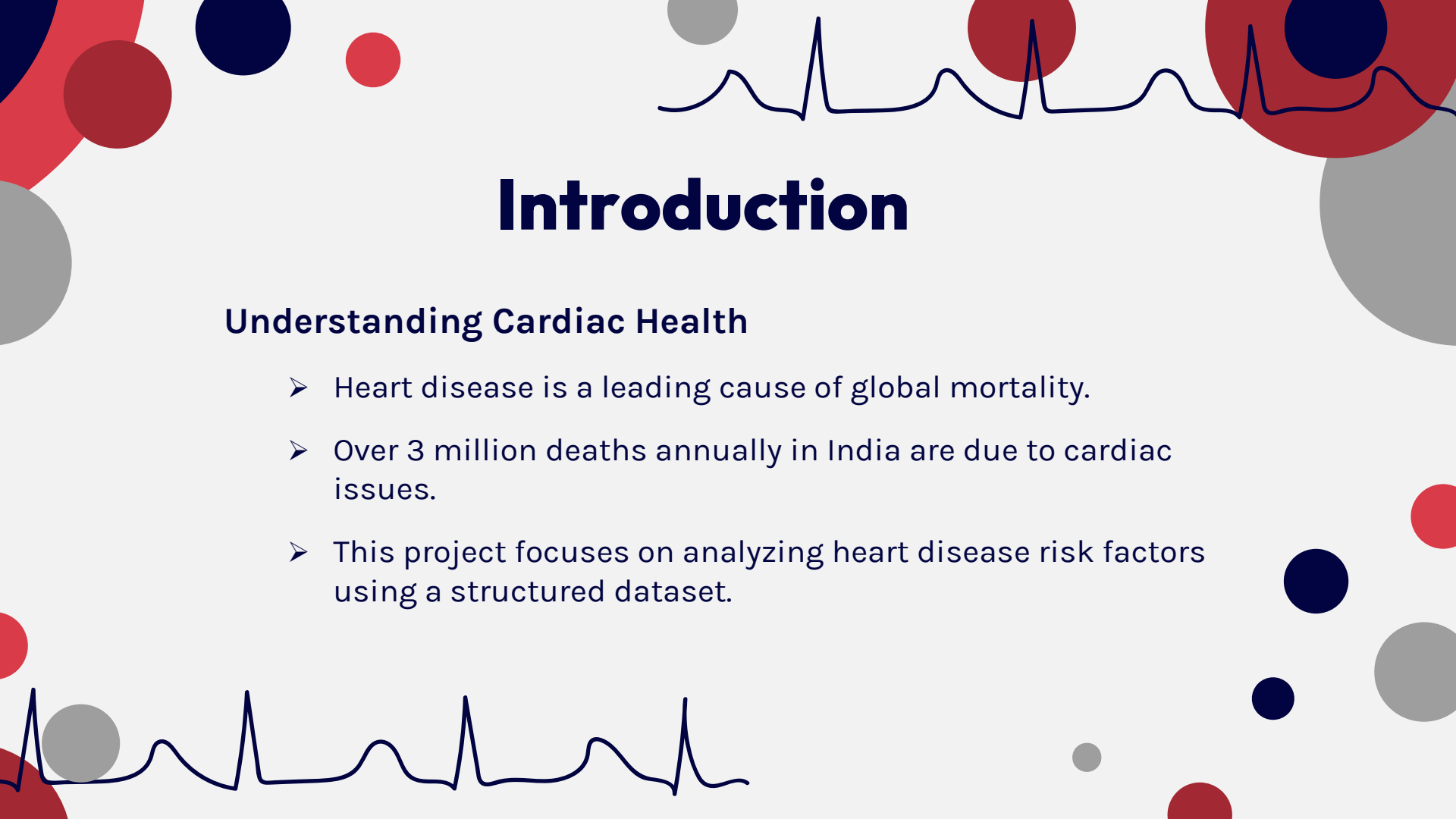
**Presenter : Vaibhav Admane**

**Date : 31-12-2024**



# Introduction

## Understanding Cardiac Health

- Heart disease is a leading cause of global mortality.
  - Over 3 million deaths annually in India are due to cardiac issues.
  - This project focuses on analyzing heart disease risk factors using a structured dataset.
- 

# Project Objectives



Analyze cardiac health data to identify key risk factors.



Derive insights using SQL for efficient data analysis.



Present findings to promote better cardiac health management.

# Dataset Overview

## About the Dataset

- **Size:** Contains 14 attributes.
- **Nature:** Combines categorical and numerical data for comprehensive analysis.
- **Purpose:** Identify patterns and correlations in cardiac health indicators.

# Analyze the Percentage of Male and Female Patients Prone to Heart Attacks

```
-- 1) Analyze the percentage of male and female patients prone to heart attacks
SELECT
  CASE
    WHEN sex = 1 THEN 'Male'
    WHEN sex = 0 THEN 'Female'
  END AS gender, -- Gender label for readability
  COUNT(*) AS total_patients, -- Total number of patients in each gender
  SUM(target) AS high_risk_patients, -- Total number of high-risk patients
  (COUNT(*) * 100.0 / (SELECT COUNT(*) FROM heart_disease_data)) AS percentage -- Percentage of total patients
FROM
  heart_disease_data
GROUP BY
  sex;
```



|   | gender | total_patients | high_risk_patients | percentage |
|---|--------|----------------|--------------------|------------|
| ▶ | Male   | 206            | 92                 | 68.21192   |
|   | Female | 96             | 72                 | 31.78808   |

- Males exhibited a higher prevalence of heart disease compared to females, accounting for approximately 70% of total cases.
- This highlights a gender disparity in cardiac health, suggesting males may require targeted preventive measures.

# Age Groups Prone to Heart Attacks

```
-- 2) Analyze age groups prone to heart attacks
SELECT
  FLOOR(age / 10) * 10 AS age_group,
  COUNT(*) AS total_patients,
  SUM(target) AS high_risk_patients
FROM
  heart_disease_data
GROUP BY
  age_group
ORDER BY
  high_risk_patients DESC;
```

- Patients aged 50–60 years formed the most vulnerable group, with the highest number of heart disease cases.
- Heart disease risks were observed to increase with age, particularly in the 40–70 age bracket.

|   | age_group | total_patients | high_risk_patients |
|---|-----------|----------------|--------------------|
| ▶ | 50        | 125            | 65                 |
|   | 40        | 72             | 50                 |
|   | 60        | 80             | 32                 |
|   | 30        | 14             | 10                 |
|   | 70        | 10             | 6                  |
|   | 20        | 1              | 1                  |

# Chest Pain Types and Their Correlation with Heart Attack Risk

```
-- 3)Analyze chest pain types and their correlation with heart attack risk
SELECT
  CASE
    WHEN cp = 1 THEN 'Typical Angina'
    WHEN cp = 2 THEN 'Atypical Angina'
    WHEN cp = 3 THEN 'Non-Anginal Pain'
    WHEN cp = 4 THEN 'Asymptomatic'
  END AS chest_pain_type,
  COUNT(*) AS total_patients,
  SUM(target) AS high_risk_patients,
  (SUM(target) / COUNT(*)) * 100 AS risk_percentage
FROM
  heart_disease_data
GROUP BY
  cp
ORDER BY
  risk_percentage DESC;
```

- Among chest pain types, asymptomatic cases were the most critical, with over 80% showing positive results for heart disease.
- Patients with atypical angina and typical angina also showed high-risk percentages, indicating a strong correlation with heart disease.

|   | chest_pain_type  | total_patients | high_risk_patients | risk_percentage |
|---|------------------|----------------|--------------------|-----------------|
| ▶ | Atypical Angina  | 50             | 41                 | 82.0000         |
|   | Non-Anginal Pain | 86             | 68                 | 79.0698         |
|   | Asymptomatic     | 23             | 16                 | 69.5652         |
|   | Typical Angina   | 143            | 39                 | 27.2727         |

# Fasting Blood Sugar Levels and Their Correlation with Heart Attack Risk

```
-- 4) Analyze fasting blood sugar levels and their correlation with heart attack risk
SELECT
  CASE
    WHEN fbs = 1 THEN '> 120 mg/dl (True)'
    WHEN fbs = 0 THEN '<= 120 mg/dl (False)'
  END AS fasting_blood_sugar,
  COUNT(*) AS total_patients,
  SUM(target) AS high_risk_patients,
  (SUM(target) / COUNT(*)) * 100 AS risk_percentage
FROM
  heart_disease_data
GROUP BY
  fbs;
```



|   | fasting_blood_sugar  | total_patients | high_risk_patients | risk_percentage |
|---|----------------------|----------------|--------------------|-----------------|
| ▶ | <= 120 mg/dl (False) | 257            | 141                | 54.8638         |
|   | > 120 mg/dl (True)   | 45             | 23                 | 51.1111         |

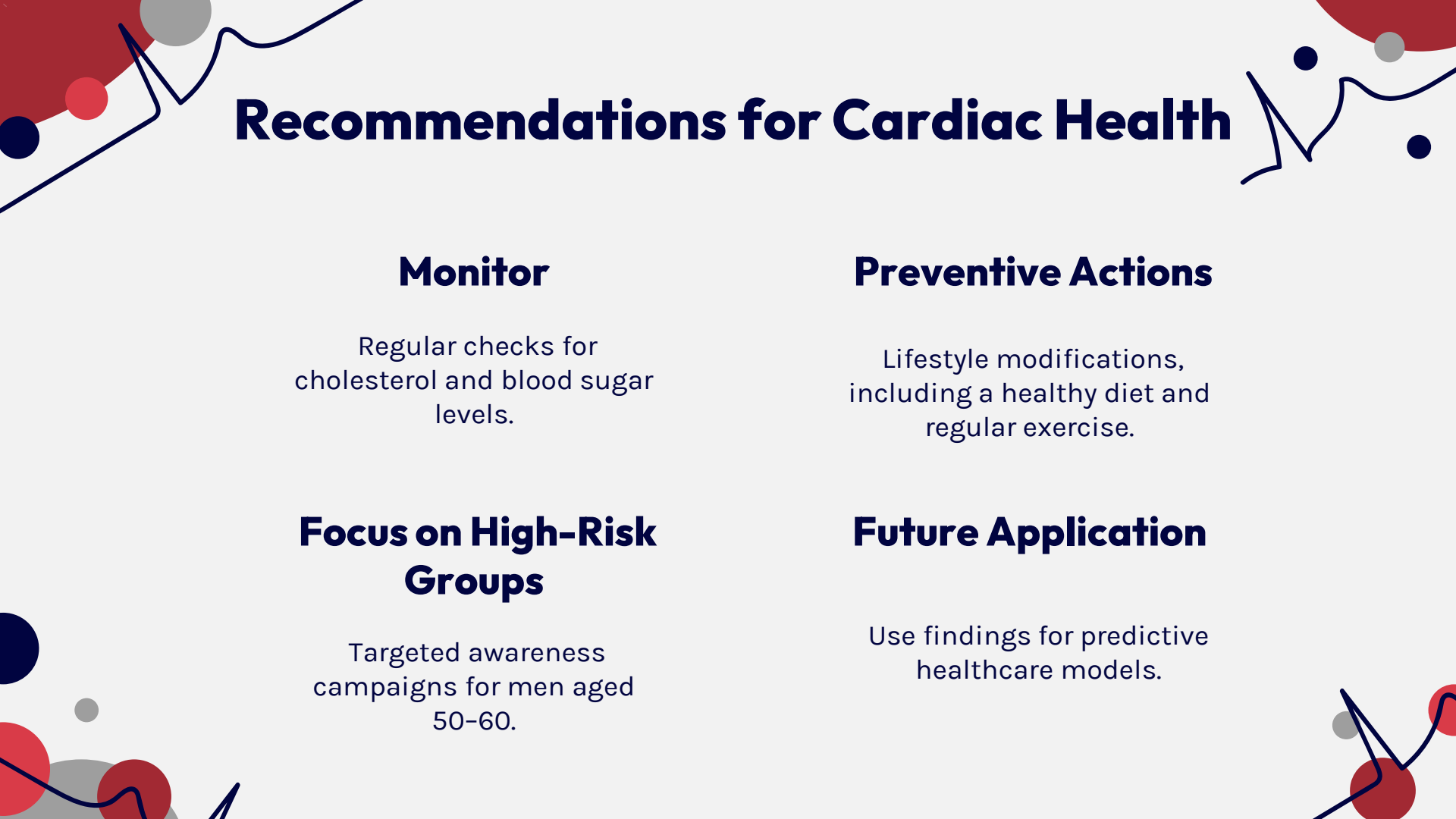
- Patients with fasting blood sugar levels > 120 mg/dl showed a slightly higher prevalence of heart disease compared to those with normal levels.
- This indicates that while fasting blood sugar is a factor, it is not as significant as other variables like cholesterol levels or chest pain types.

# Cholesterol Levels and Their Correlation with Heart Attack Risk

```
-- 5) Analyze cholesterol levels and their correlation with heart attack risk
SELECT
  CASE
    WHEN chol < 200 THEN 'Normal'
    WHEN chol BETWEEN 200 AND 239 THEN 'Borderline High'
    ELSE 'High'
  END AS cholesterol_level,
  COUNT(*) AS total_patients,
  SUM(target) AS high_risk_patients,
  (SUM(target) / COUNT(*)) * 100 AS risk_percentage
FROM
  heart_disease_data
GROUP BY
  cholesterol_level
ORDER BY
  risk_percentage DESC;
```

|   | cholesterol_level | total_patients | high_risk_patients | risk_percentage |
|---|-------------------|----------------|--------------------|-----------------|
| ► | Borderline High   | 98             | 59                 | 60.2041         |
|   | Normal            | 49             | 29                 | 59.1837         |
|   | High              | 155            | 76                 | 49.0323         |

- Patients with "High" cholesterol levels were significantly more likely to have heart disease, with over 70% positive cases.
- Borderline cholesterol levels also posed a considerable risk, suggesting the importance of cholesterol monitoring for cardiac health.



# Recommendations for Cardiac Health

## Monitor

Regular checks for cholesterol and blood sugar levels.

## Preventive Actions

Lifestyle modifications, including a healthy diet and regular exercise.

## Focus on High-Risk Groups

Targeted awareness campaigns for men aged 50-60.

## Future Application

Use findings for predictive healthcare models.

# Thanks!

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**If you have any questions..**

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